

Optimization

2.3 Successive quadratic interpolation (Powell's method)

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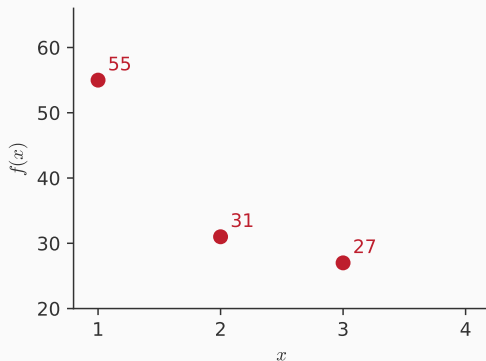
The order or the numbers?

Three evaluations of the guiding example:

$f(1) = 55$, $f(2) = 31$, $f(3) = 27$. Golden section would only use the **order**: $f(1) > f(2) > f(3)$.

But the **numbers** say more: from 1 to 2, f drops by 24; from 2 to 3, by only 4 — f is curving upwards.

Can we use the numbers to guess the shape of f ?



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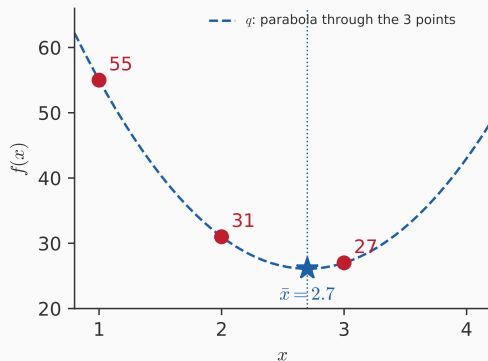
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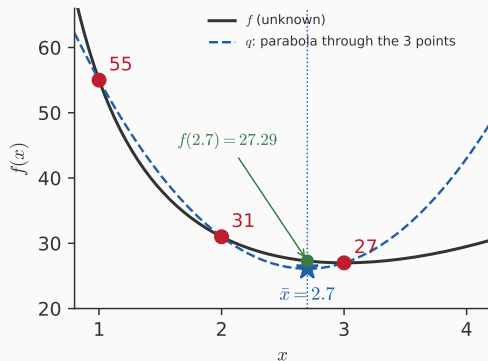
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Can we use the numbers to guess the shape of f ?

Through the three points we pass the simplest model that curves: a **parabola** q .

If q were f , the minimum would be at $\bar{x} = 2.7$.
One new evaluation: $f(2.7) = 27.29$, 0.3 away from the true minimum ($x^* = 3$).



If the parabola were f , where would the minimum be?

Parabola $q(x) = c_0 + c_1x + c_2x^2$ with $q(1) = 55$,

$q(2) = 31$, $q(3) = 27$: $q(x) = 10x^2 - 54x + 99$

(check: $10 - 54 + 99 = 55$; $40 - 108 + 99 = 31$;

$90 - 162 + 99 = 27$).

Minimum of the model: $q'(x) = c_1 + 2c_2x = 0$,

$$\boxed{\bar{x} = -\frac{c_1}{2c_2}} = \frac{54}{20} = 2.7.$$

A minimum because $c_2 = 10 > 0$ (with $c_2 < 0$ it would be a *maximum*).

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$q(2.7) = 26.1$ but $f(2.7) = 27.29$: the model is *not* f , but it points to the right region.

We keep the three best points, 2, 2.7, 3, fit a new parabola, and repeat. If f were a strictly convex quadratic, $q = f$ and $\bar{x} = x^*$ in a single iteration; near the minimum, a smooth function *looks like* a parabola.

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This is the idea of **successive quadratic interpolation** (Powell's method).

Here, "Powell's method" is this *one-dimensional* interpolation; not to be confused with Powell's *multidimensional* derivative-free conjugate-direction method (e.g. `method='Powell'` in SciPy).

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Computing $\bar{x}$ without solving a system

With any three points $x_1 < x_2 < x_3$, q is written in **Newton form** — the coefficients come out one at a time:

$$q(x) = a_0 + a_1(x - x_1) + a_2(x - x_1)(x - x_2),$$
$$a_0 = f_1, \quad a_1 = \frac{f_2 - f_1}{x_2 - x_1}, \quad a_2 = \frac{1}{x_3 - x_2} \left(\frac{f_3 - f_1}{x_3 - x_1} - a_1 \right).$$

- a_1 : slope of the chord between the first two points (first-order divided difference).
- a_2 : second-order divided difference — it measures the **curvature** ($a_2 = f''(\xi)/2$ for some ξ); it is the coefficient of x^2 , that is, $a_2 = c_2$.
 $a_2 > 0 \iff q$ has a minimum.

From $q'(\bar{x}) = a_1 + a_2(2\bar{x} - x_1 - x_2) = 0$:

$$\bar{x} = \frac{x_1 + x_2}{2} - \frac{a_1}{2a_2}$$

In the example: $a_1 = -24$, $a_2 = 10$,
 $\bar{x} = 1.5 + 1.2 = 2.7$ — the same.

Algorithm: successive quadratic interpolation

Powell: x_1 , step Δ , ε_x , ε_f

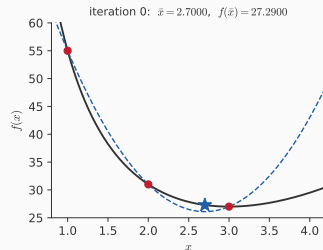
```
1:  $x_2 \leftarrow x_1 + \Delta$ ; evaluate  $f_1, f_2$ 
2: if  $f_1 > f_2$ :  $x_3 \leftarrow x_1 + 2\Delta$ ; else  $x_3 \leftarrow x_1 - \Delta$ ; evaluate  $f_3$ 
3: loop
4:   sort the three points:  $x_1 < x_2 < x_3$ 
5:    $F_{\min} \leftarrow \min(f_1, f_2, f_3)$ ;  $X_{\min} \leftarrow$  the corresponding  $x_i$ 
6:   compute  $a_1, a_2$ ; if  $a_2 \leq 0$ : reflect the worst through the
     best; repeat
7:    $\bar{x} \leftarrow \frac{x_1 + x_2}{2} - \frac{a_1}{2a_2}$ ; evaluate  $f(\bar{x})$ 
8:   if  $|F_{\min} - f(\bar{x})| \leq \varepsilon_f$  and  $|X_{\min} - \bar{x}| \leq \varepsilon_x$  then
9:     return the best of the four points
10:  end if
11:  keep the best point and the two that bracket it; if they do
     not exist, the three best
12: end loop
```

- Steps 1–2: the third point is placed on the side where f decreases, to try to *bracket* the minimum from the start.
- If x_2 is the best and is *bracketed* by two worse points, the parabola is convex and its minimum lies inside (x_1, x_3) .
- Cost: 3 initial evaluations + **1 per iteration** (+1 for each reflection).
- **Local superlinear** convergence (order ≈ 1.3 , under regularity conditions), without derivatives.

Example: $f(x) = x^2 + 54/x$, $x_1 = 1$, $\Delta = 1$

k	x_1	x_2	x_3	f_1	f_2	f_3	a_1	a_2	$\bar{x}$	$f(\bar{x})$
0	1.0000	2.0000	3.0000	55.0000	31.0000	27.0000	-24.0000	10.0000	2.7000	27.2900

Start: $x_2 = 2$; $f(1) = 55 > f(2) = 31 \Rightarrow x_3 = 1 + 2\Delta = 3$, $f(3) = 27$.

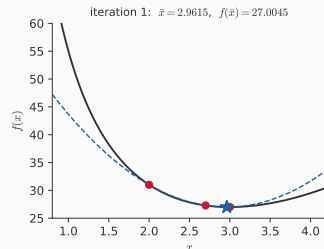


Example: $f(x) = x^2 + 54/x$, $x_1 = 1$, $\Delta = 1$

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0	1.0000	2.0000	3.0000	55.0000	31.0000	27.0000	-24.000	10.000	2.7000	27.2900
1	2.0000	2.7000	3.0000	31.0000	27.2900	27.0000	-5.300	4.333	2.9615	27.0045

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After it. 0: best point $x = 3$; with no point on its right to bracket it, the three best are kept: 2, 2.7, 3.

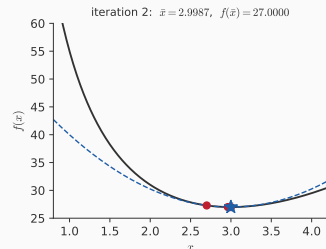


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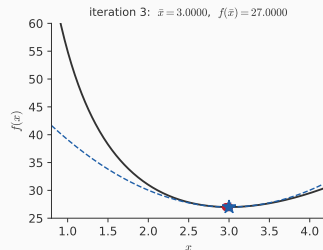
k	x_1	x_2	x_3	f_1	f_2	f_3	a_1	a_2	$\bar{x}$	$f(\bar{x})$
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2	2.7000	2.9615	3.0000	27.2900	27.0045	27.0000	-1.092	3.251	2.9987	27.0000
3	2.9615	2.9987	3.0000	27.0045	27.0000	27.0000	-0.120	3.027	3.0000	27.0000

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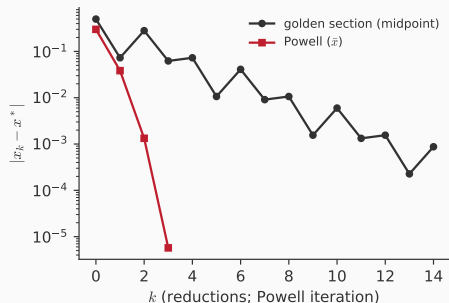
After it. 0: best point $x = 3$; with no point on its right to bracket it, the three best are kept: 2, 2.7, 3.

End (it. 3, the 4th): $|F_{\min} - f(\bar{x})| < 10^{-3}$ and $|X_{\min} - \bar{x}| < 10^{-3}$.

$x^* = 3$, $f(x^*) = 27$; 7 evaluations of f .



Golden section vs. Powell on the same problem



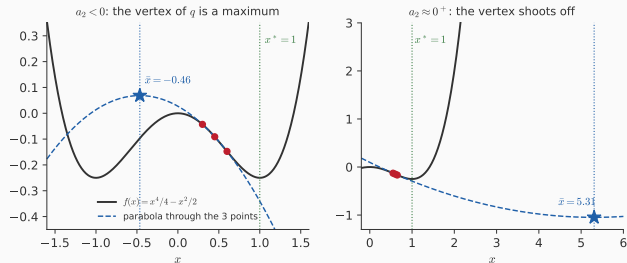
- Golden section on $[1, 4]$: interval $\times 0.618$ per reduction; the error of the midpoint oscillates and falls below 10^{-3} after 13 reductions (guarantee $L_k/2: 16$).
- Powell: error $0.3 \rightarrow 0.04 \rightarrow 0.0013 \rightarrow 6 \cdot 10^{-6}$ — 4 iterations, 7 evaluations.

But

With f unimodal, golden section always converges; quadratic interpolation may give $a_2 \leq 0$ or jump far away (next page). In practice they are combined (fminbnd).

Note — finite precision (bridge to 2.6): no method that only compares values of f can distinguish differences below $\varepsilon_m |f^*|$; here this limits $|x - x^*|$ to about $\sqrt{2\varepsilon_m |f^*| / f''(x^*)} \approx 4 \cdot 10^{-8}$.

When the parabola does not work



- $a_2 < 0$ (left, points in a concave region): q has a *maximum*; $\bar{x} = -0.46$ goes the wrong way.
- $a_2 \approx 0^+$ (right, points almost aligned near an inflection): the vertex *shoots off* to $\bar{x} = 5.31$, $f(\bar{x}) = 184$.
- Remedy: if $a_2 \leq 0$, do not compute $\bar{x}$ — **reflect the worst point through the best** (moving in the direction in which f decreases); and, as soon as possible, keep the best point **bracketed** between two worse ones: then $a_2 > 0$ and $\bar{x}$ falls inside the interval. The case of small $a_2 > 0$ is solved only by bracketing; `fminbnd` (Brent) also limits the step.

On the computer

The core of one iteration, in MATLAB:

```
[X, i] = sort(X); F = F(i);           %  $x_1 < x_2 < x_3$ 
a1 = (F(2)-F(1)) / (X(2)-X(1));
a2 = ((F(3)-F(1))/(X(3)-X(1)) - a1) / (X(3)-X(2));
if a2 <= 0           % parabola with no minimum:
    % reflect the worst point through the best and repeat
else
    xb = (X(1)+X(2))/2 - a1/(2*a2);
    fb = f(xb);           % 1 evaluation
    % stop if xb and fb change little; otherwise keep
    % the best point and the two that bracket it
end
```

The full function (PowellQuadratic, with the safeguards and info.nfev) is in the course companion code.

In the example ($x_1 = 1$, $\Delta = 1$, tolerances 10^{-3}): 4 iterations, no reflections:
 $n_f = 3 + 4 = 7$.

Safeguard: with $a_2 \leq 0$ the best point is at an end. Reflecting the worst through the best, $2X_{\min} - x_{\text{worst}}$, generates a point on the other side of the best, outside the trio, in the direction in which the observed values indicate descent.

MATLAB's fminbnd (Brent) alternates between parabolic steps and golden section steps: the speed of one, the guarantee of the other.

KEY TAKEAWAYS — 2.3

1. Golden section uses only the *order* of the values of f ; successive quadratic interpolation also uses their **magnitude**: it fits a parabola to three points and evaluates f at its minimum.
2. $\bar{x} = \frac{x_1 + x_2}{2} - \frac{a_1}{2a_2}$, with a_2 (curvature) > 0 . On a strictly convex quadratic, a single iteration.
3. Model $\rightarrow$ prediction $\rightarrow$ new evaluation: **one** evaluation per iteration and, near the minimum, locally and under suitable assumptions, superlinear convergence (order ≈ 1.3).
4. On its own it has no guarantee: $a_2 \leq 0$ or almost aligned points require safeguards (reflection, bracketing); in practice it is combined with golden section (`fminbnd`).

Next (2.4): what if we have f' ? At an interior minimum $f'(x^*) = 0$. As in order 0, we start by **comparing**: the *sign* of f' tells on which side the zero is — bisection.

References Full list: *Course bibliography*, at the end of the course PDF.

- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012 — Ch. 2.
- Fletcher, R., *Practical Methods of Optimization*, 2nd ed., Wiley, 1987 — line search by interpolation.

Resources

- Companion code: `PowellQuadratic` (safeguards, `info.nfev`). MATLAB: `fminbnd` (Brent).